Intergenerational Housing Misallocation and Fertility

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Abstract

Children increase a household's demand for living space. Yet large rental units are scarce, buying a family-sized home requires liquid wealth and access to credit, and older households own much of the stock of large homes. I study whether this allocation of housing raises the cost of having children for young households. I first develop a simple overlapping-generations model in which rental limits, down-payment constraints, and the retention decisions of older owners determine who occupies family-sized housing. I then study the same forces in a finite-life model with income risk, saving, tenure choice, transaction costs, and bequests. The model distinguishes changes in family size among current residents from changes in local births due to entry into the housing market. The quantitative model suggests that restricted access to family-sized housing matters for households deciding whether and when to have children, and that home purchases and births are closely connected near the end of the fertile years.

Status of this draft

- I study a simplified two-generation model and derive a simple analytical link between access to family-sized housing and the cost of having children. The model also identifies a potential gain from reallocating housing from older incumbents to constrained young households.
- I build a finite-life quantitative model with income risk, saving, fertility, tenure choice, transaction costs, and bequests. The solution produces a close connection between home purchases and births and places a high value on additional space among constrained renters.
- A grant for purchasing a family-sized home raises total births by approximately 3.4 percent and the housing price by 0.66 percent. Combining the grant with a tax on large homes raises total births by approximately 4.9 percent and lowers the housing price by 18.80 percent.
- The analytical section remains incomplete. The next steps are to develop the planner's problem beyond the local reallocation result, allow tenure and entry to respond in the welfare analysis, and characterize the general-equilibrium effects of the main housing policies.
- The quantitative model requires further evaluation. The current version makes a one-time choice of completed fertility, so all children arrive in a single cohort and age together. The main next extension is to add sequential fertility, alongside strengthening the old-age downsizing margin, reassessing identification, and further validating the entry response in the policy exercises.
- I do not report the project's empirical analysis in this note. Empirical moments from the CPS, ACS, and PSID are used to discipline and evaluate the quantitative model, but the construction of those moments and the reduced-form evidence are left for a subsequent version.

1 Introduction

Children increase a household's demand for living space. Yet family-sized housing is unevenly allocated across generations. Older households own much of the stock of large homes, including after their children leave, while young households deciding whether to have children are more likely to rent and have limited liquid wealth. Large rental units are scarce, and buying a family-sized home requires access to credit and a substantial down payment. This note asks whether this allocation of housing across generations raises the cost of having children and affects completed fertility—the number of children a household has by the end of its fertile years—and local births.

The mechanism operates through access to space. When the rental market does not offer sufficiently large units, a constrained renter values an additional room above its posted rent. Ownership provides access to larger homes, but a young household may be unable to finance the required down payment. At the same time, transaction costs and tax rules can encourage older owners to remain in large homes after their children leave. The resulting scarcity is therefore not only a question of the aggregate housing stock. It also concerns who occupies family-sized homes and whether those homes reach households making fertility decisions.

I study this mechanism in two steps. A simple overlapping-generations model shows how rental limits, financing constraints, and old-owner retention raise the effective cost of accommodating children. I then embed these forces in a finite-life model with income risk, saving, fertility, tenure choice, transaction costs, and bequests. The quantitative model finds that many renters making family-size decisions are constrained by the rental-size limit and that home purchases and births are closely connected near the end of the fertile years. I compare a grant for purchasing family-sized homes with a package that combines the grant with a tax on large homes. Allowing entry and population to respond, the grant raises total births by approximately 3.4 percent and the housing price by 0.66 percent; the package raises total births by approximately 4.9 percent and lowers the price by 18.80 percent.

The distinction between birth timing and completed fertility is central to this note. Quantitative lifecycle models show how economic conditions can affect both when households have children and their completed fertility (Sommer, 2016; Doepke and Kindermann, 2019). Empirical work shows that house prices, housing wealth, and mortgage access affect childbearing (Lovenheim and Mumford, 2013; Dettling and Kearney, 2014; Hacamo, 2021). Recent studies also examine longer-run outcomes: Dettling and Kearney (2025) present supplementary cohort correlations—positive for White women—between cumulative FHA/VA lending and children ever born, and van Doornik et al. (2025) use random variation in the timing of Brazilian housing-credit lotteries to estimate the effects of earlier access on completed fertility among participants observed after age 40. These findings provide the empirical motivation for a quantitative lifecycle model that separates changes in birth timing from changes in completed fertility and can be used to evaluate housing policies in equilibrium.

2 Analytical Model

2.1 Environment

The economy is populated by overlapping generations of young and old households. Each period, a mass $\bar{M}$ of potential young households is drawn from an exogenous distribution G_Y over types

$i = (y_i, b_i)$, where y_i is income and b_i is liquid wealth at entry. The potential-young pool consists of maturing children from the previous cohort and a renewal inflow. The distribution of entry wealth is taken as given rather than generated by the estates left within the model.

At the start of youth, a potential household chooses whether to live in the modeled housing market or in the outside economy. If it chooses the modeled market, it decides whether to rent or own, how much housing to occupy, its completed fertility n_i—the number of children it has by the end of its fertile years—and how much to save for old age. A household that chooses the outside economy is a nonentrant: it remains there in both periods and demands no housing in the modeled market.

Old households, indexed by j, are of two kinds. Old incumbents, last period's young owners, enter old age with financial wealth, the house they bought, and a possible accrued gain. They choose how much of the house to keep and how large an estate to leave, then exit. Old renters are last period's young renter entrants; they rent in old age and carry no capital-gains wedge.

Households have preferences over consumption, housing, and children. Children raise the household's consumption and housing needs: each child claims $\chi > 0$ units of consumption and $\kappa > 0$ units of space, so with consumption spending c_i, housing h_i, and completed fertility n_i,

$$u_i^Y = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i.$$

Old households value consumption, housing, and the bequest $e_j > 0$ they leave at exit:

$$u^2 = \log c_j^2 + \gamma \log h_j^2 + \omega_B \log e_j,$$

where $\omega_B > 0$ is the warm-glow weight. This log specialization makes the old-age allocation and the analytical results below available in closed form. The estate is a warm-glow object. The model does not assign it to a particular descendant or use it to generate the next cohort's entry wealth.

There is a fixed stock $\bar{H}$ of divisible floorspace, held by old incumbents and by passive absentee landlords: buyers purchase from either, renters occupy landlord space, and rents and sale proceeds are transfers.[1] Renting and owning are occupancy modes with different size limits, the largest sizes available to rent and to own,

$$h \leq \bar{h}^R \text{ (rent)}, \qquad h \leq \bar{h}^O \text{ (own)}, \qquad \bar{h}^R < \bar{h}^O,$$

The model represents the limited availability of large rentals by making housing sizes $h \in (\bar{h}^R, \bar{h}^O]$ available only through ownership. All floorspace trades in one market: the asset price is P, and no-arbitrage ties the rental rate r to it,

$$r = \rho P, \qquad \rho = q^{-1} - 1 + \delta + \tau^p.$$

The renter pays r as rent; the owner pays it as mortgage interest, property tax, depreciation, and the forgone return on her equity—different payments, equal per-unit value by arbitrage. The one-period bond price is q, so its net return is $q^{-1} - 1$. Given r, a higher property tax lowers P and with it the down payment on any given house.

A buyer finances at most a share $\phi \in (0, 1)$ of the purchase, so she posts a down payment $(1 - \phi)Ph$ at purchase, out of her liquid wealth. Households can save in an exogenously provided bond, but renters cannot borrow. The bond price q is exogenous.

[1] The quantitative model below allows upward-sloping supply; the fixed stock is used here to isolate the allocation of existing space.

Two taxes appear: the property tax τ^p, part of the user cost above, and the capital-gains tax. Selling a house triggers a tax: the seller pays the rate τ^g on the difference between the sale price and what she paid, in dollars. Dying instead passes the house with a stepped-up basis—the heir's purchase price is reset to the current value—so the gain is never taxed.

Where do gains come from when the real price never moves? From inflation. Dollar prices and incomes all grow at rate ϖ, so relative prices and real choices are unaffected; but the tax form compares dollars from different dates, and a house bought one period ago shows a paper gain even though its real value is unchanged. The tax on that paper gain is real,

$$\bar{\ell} = \tau^g \, \frac{P\,\varpi}{1+\varpi}$$

per unit sold, and this is the model's only contact with inflation. Every cohort carries the same gain, so every incumbent's unit bears $\bar{\ell}$ if sold while alive.[2] An estate tax, by contrast, would fall on the transfer in any form and leave the sell-or-keep margin undistorted; the gains tax distorts it precisely because death forgives it. Everything is deterministic: a buyer knows the tax her future self will face.[3]

2.2 Households

Conditional on entry, household i chooses a tenure status $m \in \{R, O\}$, housing, fertility, consumption and savings; old age is discounted at β. For $m \in \{R, O\}$,

$$\begin{aligned} W_i^m = \max_{c_i, h_i, n_i, b_i'} \; & \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i \\ & + \beta\big[\mathbf{1}\{m=R\}\, V^R(b_{j'}) + \mathbf{1}\{m=O\}\, V^O(b_{j'}, h_i)\big] \end{aligned}$$

subject to

$$\begin{gathered} c_i + r h_i + b_i' = y_i + b_i, \qquad h_i \le \bar{h}^m, \qquad b_i' \ge 0, \\ \mathbf{1}\{m=O\}\,(1-\phi) P h_i \le b_i. \end{gathered}$$

Net financial resources entering old age, after the owner housing flow cost, are $b_{j'} = b_i'/q - \mathbf{1}\{m = O\}\, r h_i$, where b_i' denotes current expenditure on the bond. Children have a market price: each child claims χ goods and κ units of space, and space rents at r, so providing for a child costs $\chi + \kappa r$ per period. A useful benchmark is equal scaling, $\chi = \kappa r/\alpha$: the child splits her cost between space and goods in the proportion $\kappa r/\chi = \alpha$, the same proportion in which her parents split their own spending.

The old incumbent enters with financial wealth b_j, housing H_j^0, and an accrued gain; old households have no income. She faces a choice with respect to her housing: sell now and pay the tax, or keep it until death, when the tax is forgiven.[4] She solves

$$\begin{aligned} V^O(b_j, H_j^0) = \max_{c_j^2, h_j^2, e_j} \; & u^2(c_j^2, h_j^2, e_j) \\ \text{subject to} \quad & c_j^2 + e_j + r\, h_j^2 = b_j + r H_j^0 - \bar{\ell}\,(H_j^0 - h_j^2), \qquad 0 \le h_j^2 \le H_j^0. \end{aligned}$$

[2] The statutory exclusion, which exempts gains below a dollar threshold, is an institutional detail I suppress; with it, small holdings escape the tax and lock-in sorts by size.

[3] Inflation generates nominal taxable gains in an otherwise stationary real environment. Coven et al. (2025) instead generate gains through real growth.

[4] The quantitative model below makes owner housing discrete, although its benchmark omits capital-gains taxation.

Selling a unit nets $r - \bar{\ell}$ in flow-equivalent units; hence the stepped-up basis acts as a subsidy to locking in. An old renter has no carried housing and no tax:

$$V^R(b_j) = \max_{c_j^2, h_j^2, e_j} \; u^2(c_j^2, h_j^2, e_j) \quad \text{subject to} \quad c_j^2 + e_j + r\, h_j^2 = b_j, \qquad 0 < h_j^2 \le \bar{h}^R,$$

and at an interior choice her marginal value of space is exactly r.

Using $P = r/\rho$, owner housing is bounded by

$$H_i^O = \min\left\{\bar{h}^O,\; \frac{b_i\,\rho}{(1-\phi)\,r}\right\}, \qquad H_i^R = \bar{h}^R.$$

H_i^m is the effective family-housing cap: a renter is capped by the stock, a buyer by the size limit or by collateral, whichever is shorter. Tenure is $W_i^Y = \max\{W_i^R, W_i^O\}$. Entry compares the inside value with an outside option $\bar{W}^E$ under extreme-value shocks of scale κ_E:

$$\pi_i^E = \frac{\exp\{W_i^Y/\kappa_E\}}{\exp\{W_i^Y/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}$$

is the probability that household i enters. The masses of entrants and nonentrants are therefore

$$M^E = \bar{M}\int \pi_i^E \,\mathrm{d}G_Y(i), \qquad M^N = \bar{M}\int (1-\pi_i^E)\,\mathrm{d}G_Y(i) = \bar{M} - M^E.$$

Only entrants contribute to housing demand in the modeled market.

2.3 Equilibrium

Aggregate demand adds young households, old incumbents, and old renters, and the single market clears:

$$\bar{M}\int \pi_i^E\, h_i \,\mathrm{d}G_Y(i) + \int h_j^2 \,\mathrm{d}F_O(j) + \int h_j^2\,\mathrm{d}F_R(j) = \bar{H},$$

where F_O and F_R are the measures of old incumbents and old renters. A stationary equilibrium consists of a rental rate r (and $P = r/\rho$), young policies and entry probabilities, old policies, and stationary measures $(\bar{M}, G_Y, F_O, F_R)$ such that:

1. the policies solve the household problems above at r;
2. the floorspace market clears, as displayed;
3. the population reproduces itself: young owners become the old incumbents, young renters the old renters, and births plus the renewal inflow recreate the cohort;

2.4 Equilibrium policies

The Lagrangian for a young agent is

$$\begin{aligned}\mathcal{L} = u_i^Y + \beta\, V_{j'} + \lambda_i\big(y_i + b_i - c_i - r h_i - b_i'\big)\\ + \mu_i\big(b_i - (1-\phi)P h_i\big) + \theta_i\big(\bar{h}^m - h_i\big) + \xi_i\, b_i',\end{aligned}$$

with $\mu_i = 0$ under rental status and $\xi_i \ge 0$ the multiplier on the no-borrowing bound $b_i' \ge 0$ for a renter. The formulas below apply at the stated interior saving and retention margins; at a corner, the corresponding complementary-slackness condition replaces the equality.

The first-order condition for saving sets the marginal value of resources today against the return tomorrow:

$$\frac{1}{c_i - \chi n_i} = \frac{\beta}{q}\,\frac{1}{c^2_{j'}} + \xi_i, \qquad \xi_i\, b'_i = 0,$$

which for savers ($\xi_i = 0$), under logarithmic old utility (warm glow $\omega_B \log e$ included) and interior old choices, collapses to a savings rule:

$$b'_i = \beta\,(1 + \gamma + \omega_B)\,(c_i - \chi n_i) + \mathbf{1}\{m = O\}\, q\bar{\ell}\, h_i,$$

with γ the old housing weight and ω_B the warm-glow weight: the household saves old age's claim on adult consumption—$\Lambda \equiv \beta(1 + \gamma + \omega_B)$—plus her discounted future tax bill, so anticipating the lock-in tax is itself a savings motive. By the same conversion, a buyer pays an effective rental price $r^O = r + q\bar{\ell}$ per unit of space—the rent plus her own discounted lock-in tax, pre-funded through savings rather than paid as a flow—while a renter pays r; write r^m for the effective price under status m. A unit of adult consumption commits $1 + \Lambda$ units of resources, one spent now and Λ carried.

Then, with no binding constraints, with lifetime resources $w_i = y_i + b_i$,

$$c_i - \chi n_i = \frac{w_i}{1 + \alpha + \vartheta + \Lambda}, \qquad n_i = \frac{\vartheta\, w_i}{(1 + \alpha + \vartheta + \Lambda)(\chi + \kappa r^m)}, \qquad h_i = \frac{\alpha\, w_i}{(1 + \alpha + \vartheta + \Lambda)\, r^m} + \kappa n_i,$$

Dividing the housing derivative of $\mathcal{L}$ by the consumption derivative, $\lambda_i = 1/(c_i - \chi n_i)$, prices every multiplier in goods. For a renter this gives

$$\frac{\alpha\,(c^R_i - \chi n^R_i)}{h^R_i - \kappa n^R_i} = r + \zeta^R_i, \qquad \zeta^R_i = \frac{\theta_i}{\lambda_i} \geq 0:$$

the left side is u_h/u_c, her marginal rate of substitution between space and consumption—the goods she would give up for one more unit of space, holding utility fixed—and it exceeds the rent exactly by the bite of the rental size limit, computable from her observed bundle. For a young owner the same two derivatives give

$$\frac{\alpha\,(c^O_i - \chi n^O_i)}{h^O_i - \kappa n^O_i} = r^O + \zeta^O_i, \qquad \zeta^O_i = \underbrace{\frac{\mu_i}{\lambda_i}(1 - \phi)P}_{\zeta^{O,F}_i,\ \text{down payment}} + \underbrace{\frac{\theta_i}{\lambda_i}}_{\text{size limit}}:$$

her marginal value of space exceeds her effective price by the bite of whichever access constraints bind—the down payment (μ_i) or the owner size limit (θ_i). When the size limit is slack, the bundle identifies the financing bite alone (F for financing): $\zeta^{O,F}_i$ is the observed ratio minus r^O. For an old incumbent at an interior retention choice,

$$\frac{\gamma c^2_j}{h^2_j} = r - \bar{\ell}:$$

she keeps marginal space she values below what the market would pay, because selling it would realize the taxable gain. Old renters carry no such wedge: their marginal value is r.

The solved policies give the level of fertility; the first-order condition behind them shows where housing enters it. Dividing that condition by the marginal utility of consumption prices the marginal child in goods: in either tenure,

$$\underbrace{\frac{\vartheta\,(c_i^m - \chi n_i^m)}{n_i^m}}_{\text{willingness to pay for a marginal child}} = \underbrace{\chi + \kappa\,(r^m + \zeta_i^m)}_{\text{full price of a child}}.$$

A child's space requirement is priced at the household's own shadow value of space, so a binding access constraint acts as a tax of $\kappa\,\zeta_i^m$ per child. Children get more expensive exactly where family-sized space is hard to reach, with market prices unchanged.

2.5 Efficiency

I first ask whether the housing frictions described above can produce an inefficient allocation of family-sized space. To isolate this question, I consider a planner that faces the housing stock, size limits, goods feasibility, and the domain restrictions, but can use lump-sum transfers to shift allocations.

Proposition 1. *Hold the stationary cross-section, entry, tenure assignments, and savings fixed; the variation below resizes two owner-occupied houses and converts no tenures. The planner's surplus from moving one unit of space from an old incumbent at an interior retention choice to a young owner whose owner size limit is slack and whose financing constraint binds ($\zeta_i^{O,F} > 0$) is*

$$\underbrace{\left(r^O + \zeta_i^{O,F}\right)}_{\textit{buyer's marginal value}} - \underbrace{(r - \bar{\ell})}_{\textit{incumbent's marginal value}} = \zeta_i^{O,F} + q\bar{\ell} + \bar{\ell} \,>\, 0.$$

Since $\zeta_i^{O,F} > 0$ and $\bar{\ell} \geq 0$, the planner can reallocate the unit to the young household and compensate the incumbent through lump-sum transfers, leaving both households weakly better off and one strictly better off. The competitive allocation therefore does not solve the planner's conditional allocation problem.

The proposition identifies a gain from moving space from an old owner to a constrained young buyer. The buyer values another unit at her effective price plus the financing gap, while the incumbent's flow-equivalent marginal value is $r - \bar{\ell}$. The common rental user cost cancels in the difference, and the tax payments are transfers rather than resource costs.

Figure 1 illustrates this comparison. The buyer's marginal value of space lies above her effective price by $\zeta^{O,F}$, while the incumbent's flow-equivalent marginal value lies below the rental user cost by $\bar{\ell}$. Compensating the incumbent at $r - \bar{\ell}$ and charging the buyer $r^O + \zeta^{O,F}$ leaves both households as well off as before and generates $\zeta^{O,F} + q\bar{\ell} + \bar{\ell}$ in surplus.

2.6 Fertility and policy

Fix a tenure status and a household whose cap binds, so that her housing is pinned at the effective family-housing cap defined in the household problem: the rental size limit for a renter, the shorter of the owner size limit and the collateral bound for a buyer. Write H for the level of that cap, $w = y + b$ for lifetime resources, and r^m for the effective price under status m. We relax the cap by a unit of space and study fertility. Concentrating the savings rule into the budget, fertility at the

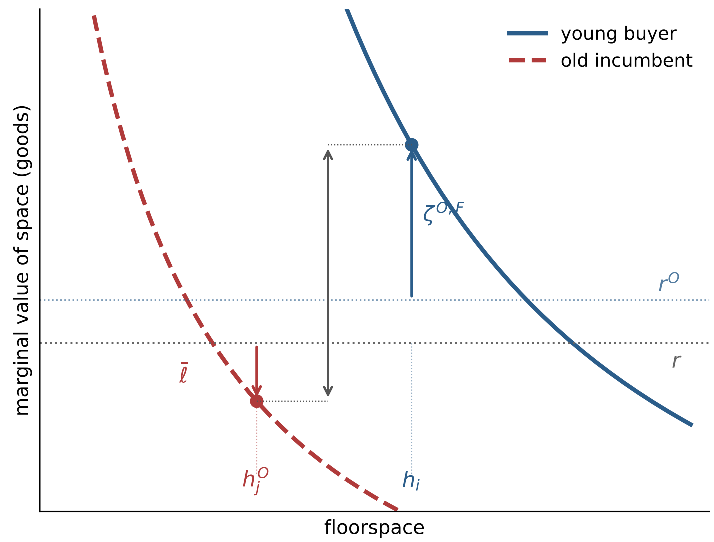

Figure 1: Misallocation. The collateral-capped buyer values marginal space at $r^O + \zeta^{O,F}$, above the rental user cost; the locked-in incumbent's flow-equivalent marginal value is $r - \bar{\ell}$, below the rental user cost; the distance between the two points is the recoverable surplus.

cap solves $\vartheta/n = (1+\Lambda)\chi/(w - r^m H - \chi n) + \alpha\kappa/(H - \kappa n)$; differentiating with respect to the cap level,

$$\frac{\mathrm{d}n}{\mathrm{d}H} = \frac{\dfrac{\alpha\kappa}{(H-\kappa n)^2} - \dfrac{(1+\Lambda)\,\chi r^m}{(w - r^m H - \chi n)^2}}{\dfrac{\vartheta}{n^2} + \dfrac{(1+\Lambda)\,\chi^2}{(w - r^m H - \chi n)^2} + \dfrac{\alpha\kappa^2}{(H-\kappa n)^2}},$$

which is positive for every binding cap if and only if $\chi \le (1+\Lambda)\,\kappa r^m/\alpha$: relaxing access raises fertility whenever the child's consumption requirement does not exceed its space requirement priced at the lifetime cost of adult consumption. The derivative races two forces: extra space makes the child's space floor κ easier to meet, pushing fertility up, while paying r^m for the space tightens the budget around the child's goods floor χ, pushing fertility down. The savings margin weakens the second force: the housing bill is paid partly by saving less, so only a fraction $1/(1+\Lambda)$ of it falls on current spending, and the threshold below which access raises fertility scales up by $(1+\Lambda)$. Figure 2 draws the relation: fertility rises along the binding cap at rate $\mathrm{d}n/\mathrm{d}H$ and flattens at h^u, where the cap stops binding.

We can then ask what effects different policies have on fertility, even without targeting it. A down-payment grant raises b_i, looser credit raises ϕ, and either change relaxes the collateral bound $b_i\rho/((1-\phi)r)$. Making larger units available to rent raises $\bar{h}^R$. These policies leave the aggregate stock $\bar{H}$ unchanged and change who can occupy it. The aggregate response depends first on how much household mass lies at the margin that the policy relaxes. Making larger units available to rent affects renters whose rental cap binds, while a down-payment grant or looser credit affects households whose access to owner housing is limited by collateral. A policy aimed at a slack constraint, or at the wrong constraint, has no direct effect. Among exposed households, the response also depends on how much the policy expands the effective family-housing cap and on how strongly fertility responds to additional space. Under equal scaling at lifetime prices, the same access gap that identifies misallocation also ranks households by their fertility response. The quantitative analysis therefore

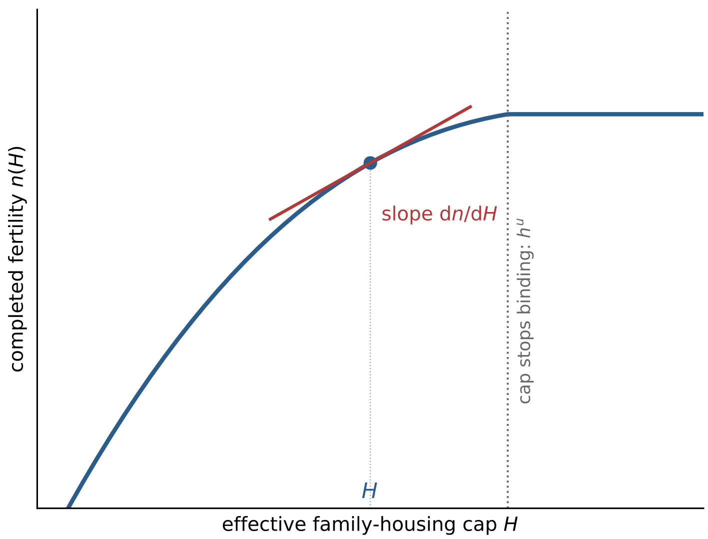

Figure 2: The fertility margin. Completed fertility against the effective family-housing cap, from the same solved case as Figure 1; the slope at the equilibrium cap is $\mathrm{d}n/\mathrm{d}H$, and the cap stops binding at h^u.

asks how much household mass lies at the relevant rental and financing margins and uses housing, wealth, and tenure moments to discipline it.

Policies can also change entry and tenure status. A change in $W_i^O - W_i^R$ moves some households across the rent–own boundary, and the gains tax affects this choice because a buyer anticipates the tax due upon a future sale. I hope to be able to say more theoretically about this soon. The quantitative model below allows these margins and the housing price to respond.

3 Quantitative Lifecycle Model

The quantitative model extends the analytical framework to a finite lifecycle. It preserves the central mechanism: children raise the demand for space, large rental units are scarce, ownership provides access to larger homes but requires a down payment, and older households may retain family-sized homes after their children leave. The model continues to have one housing market so that the allocation of space across generations, rather than spatial sorting, remains the central object.

The quantitative model adds the margins needed to evaluate this mechanism. Households face income risk, save toward homeownership, and adjust their housing over the lifecycle. Owner housing is a discrete ladder, while renters choose space continuously up to a size limit. Households choose completed fertility discretely during the fertile years; children subsequently age and leave the household, while completed fertility continues to affect bequests. Housing supply is upward sloping, and entry can change the scale of the market in counterfactuals. Older households remain in their homes because selling is costly and because housing enters the estate.[5]

[5] For now, the quantitative exercises do not study capital-gains taxation or stepped-up basis.

3.1 Lifecycle and Income

Time is discrete. Households enter adult life at age $a = 1$, retire after age J_R, and die at the end of age J. Before retirement, labor income is the product of a common wage, a deterministic age profile, and a persistent idiosyncratic component:

$$y(a, z) = \begin{cases} (1 - \tau_{\text{pay}}) w e_a z, & a \leq J_R, \\ p^{\text{ret}}, & a > J_R, \end{cases} \qquad z' \sim \Pi^z(\cdot|z), \tag{1}$$

Here, e_a is the deterministic profile of expected household labor earnings over the lifecycle, normalized to average one over working ages. The component z captures persistent differences in labor earnings across households of the same age and is normalized to have invariant mean one. The common wage w therefore sets the overall earnings scale, e_a governs the common lifecycle profile, and z governs within-age earnings risk.

I specify $\log z$ as an annual AR(1), sample the implied process every four years, and approximate it by a five-state Rouwenhorst chain. Annual persistence is 0.960 and the innovation standard deviation is 0.0645, implying four-year persistence of 0.85 while preserving the stationary log-earnings dispersion of the annual process.

3.2 Preferences and Bequests

Children raise minimum consumption and housing needs while they live at home. Let n_s be the number of dependent children and let n denote completed fertility; the two differ after children mature and leave. Household needs are

$$\begin{aligned} \bar{c}(n, s) &= \bar{c}_0 + \bar{c}_n n_s, \\ \bar{h}(n, s) &= \bar{h}_0 + \bar{h}_{\text{jump}} \mathbf{1}\{n_s \geq 1\} + \bar{h}_n n_s. \end{aligned} \tag{2}$$

The first child can create a discrete housing-space jump; additional dependent children raise the required space linearly.

Preferences are CRRA over a Stone–Geary composite of consumption and usable housing space. Write h^R for rented space and h for the owned housing stock; only one can be positive. The owner service premium applies after household needs are removed:

$$u(c, h^R, h; n, s) = \frac{\left[(c - \bar{c}(n, s))^{\alpha_c} \mathbf{h}(h^R, h; n, s)^{1-\alpha_c}\right]^{1-\sigma}}{1 - \sigma} + \psi_{\text{child}} n_s, \tag{3}$$

where

$$\mathbf{h}(h^R, h; n, s) = \begin{cases} h^R - \bar{h}(n, s), & h = 0, \\ \chi_O\left[h - \bar{h}(n, s)\right], & h > 0, \end{cases} \qquad \chi_O > 0. \tag{4}$$

When $\chi_O > 1$, ownership provides more utility from each unit of space left after household needs are met. It does not change the physical number of rooms required by children or the quantity of housing counted in market demand.

At death, households receive warm-glow bequest utility. The house passes to the estate at market value: there is no forced sale, no transaction cost, and no tax at death, so bequeathable wealth is $W^B = b + Ph$, liquid wealth plus the gross value of any house held. For $\sigma \neq 1$, bequest utility is

$$\mathcal{B}(W^B, n) = \theta_0(1 + \theta_n n) \frac{(\theta_1 + \max\{W^B, 0\})^{1-\sigma} - \theta_1^{1-\sigma}}{1 - \sigma}. \tag{5}$$

For $\sigma = 1$, the fraction is $\log(\theta_1 + \max\{W^B, 0\}) - \log \theta_1$. The normalization sets $\mathcal{B}(0, n) = 0$, while the multiplicative term makes completed fertility scale the marginal value of a positive estate. Completed fertility can therefore affect saving even after children have left the household.

3.3 Housing, Finance, and Taxes

There is one aggregate housing-services market. The rental user cost r and asset price P satisfy

$$r = (q^{-1} - 1 + \delta + \tau^p)P. \tag{6}$$

Housing supply is upward sloping:

$$H^S(r) = \bar{H} \left(\frac{r}{\bar{r}}\right)^{\eta}. \tag{7}$$

Here $\bar{r}$ is the reference rental user cost, so housing supply equals $\bar{H}$ at $r = \bar{r}$. The fixed-stock compact model is the limiting case $\eta = 0$ around the benchmark stock. More generally, η determines whether an increase in housing demand appears mainly as a higher user cost or as additional housing.

Households either rent housing services or own one of K discrete housing sizes. The owned housing stock h belongs to

$$\mathcal{H} = \{0, H_1, \ldots, H_K\}, \tag{8}$$

where $h = 0$ denotes a renter and $h = H_k$ an owner of a house of size H_k, ordered so that $H_1 < \cdots < H_K$. Thus tenure is implicit in the housing position rather than carried by a separate indicator. Renters choose h^R continuously up to $\bar{h}^R$; owners choose from the discrete ladder. The ladder is the quantitative version of the family-housing segmentation in Section 2.

Renters choose $h^R \in [0, \bar{h}^R]$ and cannot borrow. Owners choose from the discrete ladder and can borrow against the house. They pay maintenance and property taxes through $(\delta + \tau^p)PH_k$.

At origination, a buyer can finance at most a share ϕ of house value, so the required down payment is

$$(1 - \phi)PH_k. \tag{9}$$

Equivalently, adjusted liquid wealth immediately after purchase must satisfy $\tilde{b} \geq -\phi PH_k$. End-of-period saving for every owner, whether a buyer or an incumbent, faces the same debt floor:

$$b' \geq -\phi PH_k. \tag{10}$$

The down payment governs entry into ownership, while the debt floor governs resources after purchase. The distinction matters because a household can be able to service an owned home and still lack the liquid wealth needed to buy it.

Housing adjustment is costly. A sale of an owned house $h > 0$ incurs the proportional transaction cost ψ^s, so net sale proceeds are

$$\Omega(h) = (1 - \psi^s)Ph. \tag{11}$$

Transaction costs are the computed benchmark's disposal friction.[6]

[6] Capital-gains taxation and stepped-up basis are left for future quantitative work because incorporating them requires tracking each owner's tax basis.

3.4 Fertility and Child Aging

Fertility is chosen during the fertile window

$$a \in \{A_f^{\text{start}}, \ldots, A_f^{\text{end}}\}.$$

A childless household chooses

$$n' \in \mathcal{N} = \{0, 1, \ldots, \bar{n}\}. \tag{12}$$

The option $n' = 0$ means no birth in the current period. A positive choice fixes completed fertility forever and moves its children into the first dependent stage. A household that reaches the end of the fertile window without a positive choice remains childless.

In the computation, $n \in \{0, 1, 2\}$ is a completed-fertility category rather than a literal count of children.[7]

Let $s = 0$ denote no dependent children and let $s = 1, \ldots, S_c$ denote dependent-child stages. After a positive fertility choice, the household enters $s = 1$. Children age stochastically through the stages according to Π^c. While children are at home, $n_s = n$; after maturity, $n_s = 0$. Completed fertility still matters through bequest utility. All children therefore belong to a single cohort and move through the child-age stages together. The model does not yet allow sequential births or children of different ages; adding sequential fertility is the main next extension of this block.

3.5 Household Problem

A household begins the period with liquid assets b, persistent earnings component z, owned housing h, age a, completed fertility n, and child stage s. If the household is childless and in the fertile window, it first observes its fertility taste shocks and chooses family size. Conditional on the current family size, it observes its housing taste shocks and chooses the owned housing position h'. It then chooses consumption c, end-of-period liquid assets b', and, if renting, current rental services h^R.

The choice h' is made today and becomes the household's owned housing position next period; $h' = 0$ means that the household rents. Rental services h^R, by contrast, are consumed only in the current period and therefore carry no prime.

Before current housing transactions, the bond pays b/q. Liquid wealth after changing housing from h to h' is

$$\tilde{b}(b, h, h') = \frac{b}{q} + \mathbf{1}\{h > 0,\ h' \neq h\}\Omega(h) - \mathbf{1}\{h' > 0,\ h' \neq h\}Ph'. \tag{13}$$

When $h' = h$, no house is traded. An owner who changes housing receives the net sale proceeds $\Omega(h)$, while a household choosing a new owned house pays Ph'. The renter and owner problems below are therefore evaluated at $\tilde{b}(b, h, 0)$ and $\tilde{b}(b, h, h')$, respectively.

Let the continuation value after earnings risk and child aging be

$$\bar{V}(b', z, h', a + 1, n, s) = \mathbb{E}\left[V(b', z', h', a + 1, n, s') \mid z, s, n\right]. \tag{14}$$

The expectation is over next period's earnings component and child stage. The terminal value is

$$V(b, z, h, J + 1, n, s) = \mathcal{B}(b + Ph, n).$$

[7] The category $n = 1$ represents families with one or two children, while $n = 2$ represents families with three or more. The maintained scaling assigns two children to each unit of the index, so the completed-fertility measure matched to the CPS is reported as $2\mathbb{E}[n]$. Preferences, child needs, bequests, and demographic flows are evaluated in index units.

Conditional on renting, the branch value is

$$
\begin{aligned}
V^R(\tilde{b}, z, a, n, s) = \max_{c, h^R, b'} \quad & u(c, h^R, 0; n, s) + \beta \bar{V}(b', z, 0, a+1, n, s) \\
\text{s.t.} \quad & c + r h^R + b' = \tilde{b} + y(a, z), \\
& 0 \le h^R \le \bar{h}^R, \qquad b' \ge 0.
\end{aligned}
\tag{15}
$$

Conditional on choosing an owned house $h' > 0$ after any adjustment, the branch value is

$$
\begin{aligned}
V^O(\tilde{b}, z, h', a, n, s) = \max_{c, b'} \quad & u(c, 0, h'; n, s) + \beta \bar{V}(b', z, h', a+1, n, s) \\
\text{s.t.} \quad & c + (\delta + \tau^p) P h' + b' = \tilde{b} + y(a, z), \\
& b' \ge -\phi P h'.
\end{aligned}
\tag{16}
$$

The debt floor is the same for a buyer and an incumbent. Renting offers a continuous choice of space but no borrowing; ownership offers the larger discrete menu and borrowing against the house, at the cost of the initial down payment and future adjustment costs. The housing-position value for a feasible $h' \in \mathcal{H}$ is

$$
\widetilde{V}^H(h' | b, z, h, a, n, s) = \begin{cases} V^R(\tilde{b}(b, h, 0), z, a, n, s), & h' = 0, \\ V^O(\tilde{b}(b, h, h'), z, h', a, n, s), & h' \in \mathcal{H} \setminus \{0\}. \end{cases} \tag{17}
$$

The rent-or-own and owner-size choices have Type-I extreme-value shocks with scale κ_T:

$$
\pi^H(h' | b, z, h, a, n, s) = \frac{\exp\{\widetilde{V}^H(h' | b, z, h, a, n, s)/\kappa_T\}}{\sum_{\hat{h} \in \mathcal{H}} \exp\{\widetilde{V}^H(\hat{h} | b, z, h, a, n, s)/\kappa_T\}}, \tag{18}
$$

and the inclusive housing value is

$$
V^H(b, z, h, a, n, s) = \kappa_T \log \sum_{\hat{h} \in \mathcal{H}} \exp\{\widetilde{V}^H(\hat{h} | b, z, h, a, n, s)/\kappa_T\}. \tag{19}
$$

For a childless household in the fertile window, the value of fertility option n' is

$$
\widetilde{V}^F(n' | b, z, h, a, 0, 0) = V^H(b, z, h, a, n', s(n')), \qquad s(n') = \mathbf{1}\{n' > 0\}. \tag{20}
$$

With Type-I extreme-value shocks of scale κ_F,

$$
\pi^F(n' | b, z, h, a, 0, 0) = \frac{\exp\{\widetilde{V}^F(n' | b, z, h, a, 0, 0)/\kappa_F\}}{\sum_{\hat{n} \in \mathcal{N}} \exp\{\widetilde{V}^F(\hat{n} | b, z, h, a, 0, 0)/\kappa_F\}}. \tag{21}
$$

The value before the fertility draw is the corresponding inclusive value:

$$
V(b, z, h, a, 0, 0) = \kappa_F \log \sum_{\hat{n} \in \mathcal{N}} \exp\{\widetilde{V}^F(\hat{n} | b, z, h, a, 0, 0)/\kappa_F\}. \tag{22}
$$

Outside the fertile window, or after a positive fertility choice, this stage is degenerate and $V(b, z, h, a, n, s) = V^H(b, z, h, a, n, s)$. The family-size decision therefore incorporates the housing position chosen in the same period. A birth can change current space and tenure immediately, not only the household's continuation value.

3.6 The Entry Problem

At age 18, a potential household chooses whether to live in the modeled housing market or in the outside economy. This choice is made before initial liquid wealth and the persistent earnings component are realized. Conditional on entry, the household draws (b, z) from G_E and begins adult life as a childless renter. Locally born and outside-origin households draw from the same distribution G_E.

The value of entry is therefore

$$W^E(r) = \int V(b, z, 0, 1, 0, 0; r)\, \mathrm{d}G_E(b, z).$$

The outside economy provides value $\bar{W}^E$. With Type-I extreme-value tastes of scale κ_E, the fraction of potential households that enter is

$$q^E(r) = \frac{\exp\{W^E(r)/\kappa_E\}}{\exp\{W^E(r)/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}. \tag{23}$$

Housing costs affect entry through $W^E(r)$: a lower cost of living in the modeled market raises the fraction of potential households that enter.

3.7 Entry Equilibrium and Stationary Scale

Let g denote the beginning-of-period composition of adult households over liquid wealth, the persistent earnings component, inherited owned housing, age, family size, and child age, (b, z, h, a, n, s), normalized to integrate to one. Let S denote the total adult population.

There are two sources of potential entrants. A fixed mass M comes from the outside economy. In addition, locally born children mature into potential entrants. Let $B_0(g, r)$ denote the flow of mature locally born potential entrants per unit of adult population. The total potential-entrant pool is therefore $M + SB_0(g, r)$.

Let $E_0(g, r)$ denote the inflow of young adults required per unit of adult population to reproduce the stationary composition g—equivalently, the mass of age-one households in g. Stationarity requires the number who enter to equal the number of young adults needed to replace the population:

$$q^E(r)\,[M + SB_0(g, r)] = SE_0(g, r). \tag{24}$$

The left-hand side is the mass entering from the two potential-entrant pools; the right-hand side is the inflow required to sustain an adult population of size S.

For $M > 0$, a finite positive adult population requires $E_0(g, r) > q^E(r)B_0(g, r)$. Under this condition, rearranging (24) gives

$$S = \frac{q^E(r)M}{E_0(g, r) - q^E(r)B_0(g, r)}. \tag{25}$$

Housing costs affect population scale through the entry probability $q^E(r)$, while fertility affects scale through the future flow $B_0(g, r)$ of locally born potential entrants.

The benchmark normalizes $S^* = 1$.[8] Given the benchmark composition, I choose $\bar{W}^E$ to deliver the

[8] Counterfactuals hold $(M, \bar{W}^E, \kappa_E, G_E)$ fixed while entry, population scale, and the housing-market equilibrium adjust.

benchmark entry probability $q^{E,*}$ and set

$$M = \frac{E_0^*}{q^{E,*}} - B_0^*.$$

3.8 Stationary Equilibrium

Throughout the equilibrium definition, q, δ, and τ^p are fixed, so $P = r/\rho$.

Definition 1 (Stationary equilibrium)**.** A stationary equilibrium is a rental user cost and asset price (r, P), value functions and policies, housing and fertility choice probabilities, an entry probability, stationary composition g, adult scale S, and aggregate housing quantities such that:

1. households solve the branch problems (15)–(16), housing choice is given by (18), fertility choice is given by (21), and terminal utility is (5);
2. household choices and saving, together with earnings risk, child and adult aging, death, and an inflow $E_0(g, r)\,\mathrm{d}G_E(b, z)$ of age-one childless renters, define a normalized transition operator $\mathcal{T}(g; r, S)$. Stationarity requires
$$g = \mathcal{T}(g; r, S); \tag{26}$$
3. entry and population scale satisfy (23) and (24), with finite scale given by (25);
4. let $\Pi^H(h' \mid b, z, h, a, n, s)$ and $\widehat{h}^R(b, z, h, a, n, s)$ denote the housing-position probability and expected rental space conditional on renting, respectively, after integrating over the current family-size choice. Per-unit physical housing demand is
$$\begin{aligned} \widehat{H}^D(g, r) = \int \Bigg[& \Pi^H(0 \mid b, z, h, a, n, s)\widehat{h}^R(b, z, h, a, n, s) \\ & + \sum_{k=1}^{K} \Pi^H(H_k \mid b, z, h, a, n, s) H_k \Bigg] \mathrm{d}g(b, z, h, a, n, s). \end{aligned} \tag{27}$$
5. the housing market clears in levels and the user-cost relation holds:
$$S\widehat{H}^D(g, r) = H^S(r), \qquad r = (q^{-1} - 1 + \delta + \tau^p)P; \tag{28}$$
6. taxes, pensions, and any rebates satisfy the government accounting rule specified for the calibration or counterfactual.

Equilibrium links family formation to housing through two margins. Children change housing demand within a household, and, when locally born children feed the future entrant pool, they also change the scale of the market. Housing supply determines how much of either margin is absorbed by quantities rather than by the user cost.

3.9 Computation

One model period is four years. Period $j = 0, \ldots, 16$ begins at age $18 + 4j$; the last interval covers ages 82–85. Retirement begins at 66, and the last period in which a birth can occur begins at 42. Annual parameters are converted before solution:

$$q = (1 + i_{\text{ann}})^{-4}, \quad \beta = \beta_{\text{ann}}^4, \quad \delta = 1 - (1 - \delta_{\text{ann}})^4, \quad \tau^p = 4\tau_{\text{ann}}^p.$$

The property-tax conversion is the undiscounted sum of four annual flow rates. Income and consumption-flow quantities are expressed in four-year units; housing remains in rooms.

The liquid-wealth grid spans $[-12, 30]$, with a dense core on $[-5, 7]$ and a thinner upper tail beyond 15. The five-state Rouwenhorst process has the following grid for the earnings component: $z \in \{0.613, 0.773, 0.974, 1.227, 1.546\}$ and invariant weights $\pi_z = (0.0625, 0.25, 0.375, 0.25, 0.0625)$. One dependent-child stage has an expected duration of 18 years, giving a four-year maturity probability of 2/9. With deterministic J-period lives, the stationary normalized composition has age-one mass $E_0(g, r) = 1/J$. The balanced pension equals payroll-tax revenue per retiree in the stationary age distribution.

I solve the household problem by backward induction and obtain the stationary distribution by forward iteration. The user cost adjusts until aggregate housing demand equals supply.

4 Calibration

I organize the model parameters into two groups. The first consists of parameters assigned from external evidence, measured directly in the data, or fixed as maintained restrictions. The second consists of parameters estimated internally by matching model moments to their empirical counterparts. I describe the two groups in turn, explain which moments are particularly informative about the internally estimated parameters, and then report the model's fit.

4.1 Parameters Set Outside the Estimation

I first describe the parameters assigned outside the estimation. Table 1 summarizes their values and sources.

Demographics and preferences. A model period is four years. Households enter adult life at age 18, retire at age 66, and live at most through age 85. Survival is certain before retirement and thereafter follows four-year probabilities constructed from the combined male and female survivor counts in the 2023 Social Security period life table. Dependent children remain in the household for an expected 18 years, implying a maturity probability of 2/9 per period. I set risk aversion to $\sigma = 2$.

Housing finance and transaction costs. Following Greaney et al. (2025), I set the annual real return on liquid wealth to $i_{\text{ann}} = 0.02$, annual housing depreciation to $\delta_{\text{ann}} = 0.011$, the proportional selling cost to $\psi^s = 0.06$, and the labor-income tax rate to $\tau_{\text{pay}} = 0.179$. The return is based on the average ten-year real interest rate, depreciation combines BEA residential-structure depreciation with an adjustment for the land share of house values, and the tax rate comes from OECD data. I set the annual property-tax rate to $\tau^p_{\text{ann}} = 0.01$. Mortgage debt cannot exceed a share $\phi = 0.80$ of the value of the house, so a purchase requires at least 20 percent equity. The benchmark imposes no separate payment-to-income limit.

Income process. I impose a deterministic lifecycle profile of household earnings, e_a, normalized to average one over working ages. Within age, log income follows an annual AR(1) process with persistence $\rho_z = 0.960$, which I sample at the model's four-year frequency and approximate with a five-state Rouwenhorst chain. This persistence is near the upper end of the PSID-based estimates summarized by Sommer and Sullivan (2018). I set the annual innovation standard deviation to $\sigma_z = 0.0645$, below estimates of gross income risk in that literature. I treat this variance as provisional rather than as an externally validated estimate of income risk.[9] The quantitative results are conditional on this restriction.

Housing products and supply. Housing is indivisible and available in a limited assortment of types and sizes (Davis and Van Nieuwerburgh, 2015). I represent this lumpiness with five owner-occupied house sizes, $\{H_k\}_{k=1}^{5} = \{2, 4, 6, 8, 10\}$ rooms. Renters choose housing continuously up to $\bar{h}^R = 6$ rooms, the 90th percentile of the rental-room distribution in the matched ACS sample. These menus summarize the observed tenure-specific distribution of rooms. I set the housing-supply elasticity to $\eta = 1.75$, following the population-weighted metropolitan estimate in Saiz (2010).

Entry wealth and bequests. Households enter at age 18 with a draw from the PSID net-worth-to-income distribution of childless renters aged 18–24. I impose $\theta_n = 0$, so bequest utility does not vary directly with the number of children. Among PSID households aged 65–75, the median total-wealth-to-income ratio for households with two or more children exceeds that for one-child households by 0.101. Children affect estate accumulation through their effects on housing needs, tenure, and saving.

[9] With innovation risk in the empirical range, households in the lowest persistent-income states cannot always finance the model's minimum consumption and housing bundle. This conflict may reflect missing insurance mechanisms—the current model has no means-tested safety net, family transfers, or labor-supply margin—but it may also indicate that the hard minimum requirements implied by the Stone–Geary specification are too restrictive. Future revisions will therefore reconsider both the missing insurance margins and the Stone–Geary preference structure.

Table 1: Parameters set outside the estimation

Parameter	Value	Source or restriction
Model period	4 years	Model timing
Entry, retirement, and maximum ages	18, 66, and 85	Model timing
Post-retirement survival	Age-specific	2023 Social Security period life table
Expected years with dependent children	18 years	Child-maturity restriction
Risk aversion σ	2	
Annual real return i_{ann}	2.0%	Average 10-year real rate, 1962–2024; Greaney et al. (2025)
Annual housing depreciation δ_{ann}	1.1%	BEA structures depreciation, adjusted for the residential land share; Greaney et al. (2025)
Annual property tax τ^p_{ann}	1.0%	Maintained tax rate
Financed share ϕ	0.80	20% down-payment restriction; Davis and Van Nieuwerburgh (2015)
Sale cost ψ^s	6%	Davis and Van Nieuwerburgh (2015); Greaney et al. (2025)
Labor-income tax τ_{pay}	17.9%	OECD U.S. average; Greaney et al. (2025)
Income persistence ρ_z	0.960 annually	see text
Income innovation standard deviation σ_z	0.0645 annually	see text
Owner house sizes $\{H_k\}$	$\{2, 4, 6, 8, 10\}$ rooms	Occupied-room distribution in the ACS; see text
Largest rental unit $\bar{h}^R$	6 rooms	Matched ACS rental-size distribution
Housing-supply elasticity η	1.75	Saiz (2010)
Entrant wealth distribution	Wealth-to-income distribution	Childless renters aged 18–24 in the PSID

4.2 Internally Estimated Parameters

I estimate the remaining 14 parameters internally using 15 moments describing completed fertility, childlessness, housing consumption, tenure, wealth accumulation, and aggregate occupied housing. Table 2 reports the estimates. Although the parameters are chosen jointly, I organize the discussion into economic blocks and describe which moments are most informative about each block.

Table 2: Internally estimated parameters

Parameter	Interpretation	Estimate
β_{annual}	Annual discount factor	0.9912
α_c	Consumption weight	0.5912
$\bar{c}_0$	Baseline consumption requirement	1.2596
$\bar{c}_n$	Consumption cost per dependent child	0.3929
$\bar{h}_0$	Baseline housing requirement	0.3915
$\bar{h}_{\text{jump}}$	Housing-need increase with the first child	1.6021
$\bar{h}_n$	Housing requirement per dependent child	0.1644
ψ_{child}	Utility from children	0.1979
κ_F	Dispersion in family-size choices	2.0434
χ_O	Utility premium from owner-occupied housing	1.1133
$\bar{H}$	Housing-supply scale	8.6456
θ_0	Strength of the bequest motive	0.3118
θ_1	Bequest curvature shift	0.3973
κ_T	Dispersion in tenure choices	0.0100

Saving. The annual discount factor β_{annual} governs households' willingness to carry resources across ages. Liquid wealth among young childless renters, the older nonhousing-wealth share, and the late-life estate median are particularly informative about it.

Housing demand, tenure, and supply. I use housing-size differences by family composition to discipline the housing requirements. Mean rooms among childless renters is particularly informative about $\bar{h}_0$, the increase associated with the first child disciplines $\bar{h}_{\text{jump}}$, and the difference between larger and smaller families disciplines $\bar{h}_n$. The owner–renter rooms difference and the share of childless owners in large homes help discipline α_c and χ_O. Ownership rates by age and family status are particularly informative about χ_O and κ_T, while aggregate occupied rooms disciplines the housing-supply scale $\bar{H}$.

Consumption needs and family size. The consumption requirement is $\bar{c}_0 + \bar{c}_n n_s$. Outcomes for childless households, including young liquid wealth, are particularly informative about the baseline requirement $\bar{c}_0$, while fertility and differences in housing and tenure by family composition help discipline the incremental cost $\bar{c}_n$. Completed fertility and childlessness jointly discipline these costs with the direct utility value of children ψ_{child} and the dispersion of family-size choices κ_F. These parameters are estimated jointly because the same family outcomes respond to both the cost and the utility value of children.

Bequests. The bequest parameters θ_0 and θ_1 govern the strength and wealth dependence of the bequest motive. The late-life median estate is particularly informative about θ_0, while the older nonhousing-wealth share, together with old-age ownership, provides additional information about θ_1. These moments also respond to saving and tenure choices, so the two bequest parameters are disciplined jointly with those blocks.

4.3 Target Moments and Fit

The target system combines completed fertility and childlessness from the CPS, tenure and housing-size moments by age and family status from the ACS, housing responses to changes in the number of children, and wealth moments from the PSID. Table 3 reports all 15 target moments and their model counterparts.

Table 3: Target moments and model fit

Moment	Data	Model
Family size and family housing		
Completed fertility	1.918	2.034
Childlessness rate	0.188	0.189
Family ownership gap	0.168	0.219
Housing increment with the first child, rooms	0.664	0.681
Rooms gap, 3+ versus 1–2 children, ages 30–55	0.368	0.607
Tenure and housing consumption		
Ownership rate, ages 30–55	0.575	0.658
Renter mean rooms, no children at home, ages 30–55	3.805	3.963
Owner share with rooms ≥ 6, no children at home, ages 30–55	0.596	0.760
Owner–renter mean rooms gap, no children at home, ages 30–55	2.419	2.323
Ownership rate, ages 25–34	0.341	0.274
Ownership rate, ages 65–75	0.764	0.954
Mean occupied rooms, ages 18–85	5.780	5.673
Wealth and bequests		
Young childless-renter liquid wealth / annual income, ages 25–35	0.179	0.328
Median total estate wealth / annual income, ages 76–84	6.501	6.431
Share with nonhousing wealth $\geq$ annual income, ages 65–75	0.608	0.610

The model closely matches childlessness, the housing increment associated with the first child, renter mean rooms, the owner–renter rooms difference, aggregate occupied rooms, the late-life median estate, and the older nonhousing-wealth share. It produces completed fertility of 2.034, compared with 1.918 in the data. The model overstates the overall ownership rate, the family ownership gap, the share of childless owners in large homes, and the rooms difference between larger and smaller families, while understating ownership at ages 25–34.

The wealth moments reveal a lifecycle tension. The two late-life wealth targets are closely matched, but young childless renters accumulate liquid wealth equal to 0.328 of annual income, compared with 0.179 in the PSID. The estimated annual discount factor of 0.9912 should therefore be read as a joint calibration outcome rather than as a direct estimate of household patience.

The largest miss is old-age ownership: the model produces an ownership rate of 0.954 at ages 65–75, compared with 0.764 in the ACS. The model has no health or long-term-care expense risk and no age-dependent maintenance burden that induces older owners to leave homeownership. Together with transaction costs and a bequest motive that values the house at death, this makes owner housing unusually persistent late in life. The untargeted estate distribution shows a related limitation: the model produces an estate p90–p50 ratio of 1.75, compared with 3.45 in the PSID, and understates nonhousing wealth in the upper part of the late-life distribution. I therefore do not use the current calibration to draw conclusions that depend on the late-life ownership path or portfolio composition.

The calibration remains provisional because the income variance and the late-life exit and portfolio margins require further work. The quantitative exercises that follow focus on family housing needs, tenure, and the down-payment constraint and should be read conditional on these limitations.

5 Lifecycle Housing Patterns

Figure 3 compares the model's lifecycle housing and child-at-home profiles with the ACS. The model understates homeownership through the mid-thirties and overstates it from the early forties onward. Mean rooms are below the data throughout the lifecycle. The child-at-home profile has the right hump shape but is too delayed and persistent: the model peaks at 0.522 at age 42, whereas the ACS profile peaks at 0.646 at age 41.

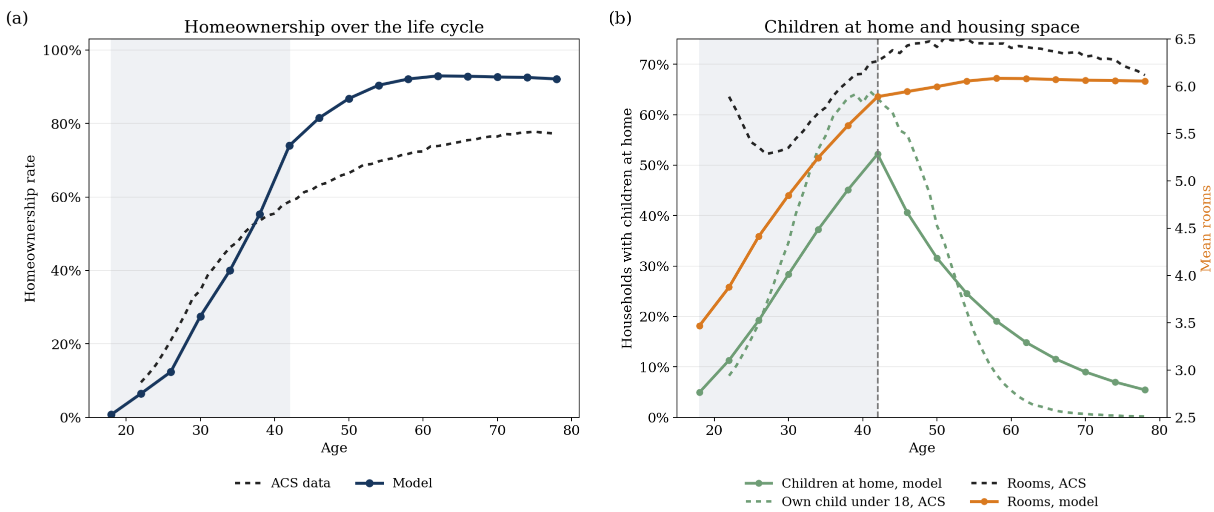

Figure 3: Lifecycle ownership, children at home, and housing consumption. Panel (a) compares model homeownership with the ACS household-head profile. Panel (b) compares the model share of households with dependent children at home with the ACS share of household heads with an own child under age 18, and compares mean rooms in the model and the ACS. The shaded region marks ages 18–42, the family-size-choice window in the model.

At age 38 the model child-at-home share is 18.2 percentage points below the ACS; at age 62 it is 11.7 points above it. Thus the discrepancy is not completed fertility "at" a young age. It is a timing and duration mismatch in the stock of dependent children at home, the object that directly shifts household space requirements in the model.

Figure 4 fixes households at age 42 as childless renters and varies liquid wealth and income. Both choices are strongly nonlinear. Higher-income households select children and larger homes at lower levels of liquid wealth, while the steps in physical rooms reflect movements through the discrete owner ladder.

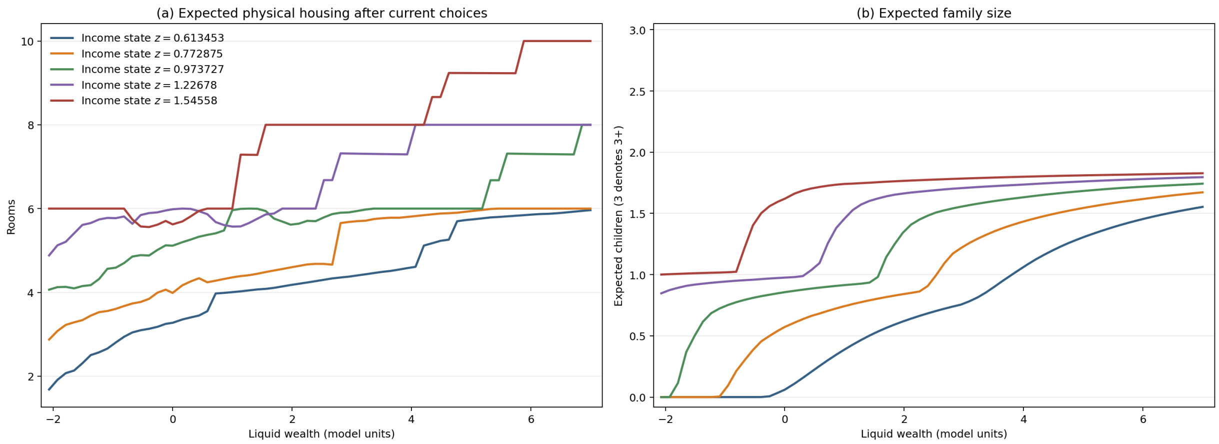

Figure 4: Housing and family-size policies at the end of the fertile window. Panel (a) reports expected physical rooms after integrating over the current family-size choice and the housing choice that follows it. Panel (b) reports expected family size. Each line corresponds to a current income state.

The housing rule is not monotone at every tenure transition: for some states, crossing into ownership reduces physical rooms. This can occur because owner housing is discrete and carries an ownership premium.

6 Quantitative Evidence on the Mechanism

I use the household's model-implied marginal rate of substitution to measure the value of relaxing the rental-size constraint. For a fertile childless renter state i, let (c_i^B, h_i^B) denote consumption and housing on the first positive family-size branch conditional on continuing to rent, and define usable consumption and space by $\widetilde{c}_i = c_i^B - \bar{c}(1)$ and $\widetilde{h}_i = h_i^B - \bar{h}(1)$. If this branch reaches the rental cap, the excess marginal value of another room is

$$\zeta_i^R = \frac{1-\alpha_c}{\alpha_c}\frac{\widetilde{c}_i}{\widetilde{h}_i} - r,$$

where r is the rental user cost per room. Thus $r + \zeta_i^R$ is the household's local marginal willingness to substitute consumption for space. This is a model-implied marginal valuation, not an observed bid or the welfare gain from providing an entire additional room.

Among current fertile renters whose conditional positive-family renter branch reaches the cap, the stationary-mass-weighted mean of ζ_i^R is 0.178 goods units. This is the value above the 0.130 rental user cost. Their total marginal value of another unit of space is 0.307 goods units, or 2.37 times its user cost.

Table 4 compares who owns homes with at least six rooms in the model and in the ACS.

Table 4: Who holds the family-sized owner stock (rooms ≥ 6)

Holder	Data (ACS)	Model
Age 18–29	3.4%	2.4%
Age 30–45	24.8%	16.9%
Age 46–64	43.3%	43.5%
Age 65+	28.5%	37.2%
Held by households 46+	71.8%	80.7%
Held by households with no children at home	54.9%	74.8%

In the ACS, households past the fertile years hold 72 percent of family-sized owner housing, and households with no children at home hold 55 percent. The corresponding model shares are 81 and 75 percent. These quantities describe who holds the stock; whether that allocation is inefficient depends on the holders' and potential occupants' marginal values of space.

7 Housing Policy

I use these preliminary policy exercises to ask how relaxing access to family-sized housing affects fertility. The first is a purchase grant for renters with dependent children who buy a home with at least six rooms. The grant covers roughly one-third of the required down payment on an eligible family-sized home. The second exercise combines the same grant with an increase in the annual property-tax rate on owner-occupied housing from 1 to 2 percent. House prices, market entry, and population adjust in both exercises.

The purchase grant raises total births by approximately 3.4 percent. The property-tax-plus-grant package raises total births by approximately 4.9 percent. The corresponding housing-price changes are +0.66 percent and −18.80 percent, respectively. Table 5 summarizes the results.

Table 5: Population-adjusted effects of housing policy

Policy	Fertility effect: Δ total births	Δ housing price
Purchase grant for family-sized homes	+3.4%	+0.66%
Package: higher property tax and purchase grant	+4.9%	−18.80%

The total-birth effects combine changes in family formation among incumbent households with changes in the number of households living in the market. The grant's approximately 3.4 percent increase in births comes with a 0.66 percent increase in the housing price. The package's approximately 4.9 percent increase in births comes with an 18.80 percent decline in the housing price.

These exercises are most useful for comparing mechanisms. The model has too little old-age downsizing, omits the capital-gains basis channel, and generates the distorted family-formation timing documented above. Because the model represents one housing market, the total-birth measure cannot distinguish newly created births from births relocated from other markets.

8 Conclusion

Children raise demand for space, but young households face a rental-size limit and, upon buying, a down payment. At the same time, older households hold much of the family-sized owner stock. The analytical model shows how these features raise the effective cost of accommodating children and can misallocate housing across generations.

The population-adjusted counterfactuals provide the main quantitative result. The purchase grant raises total births by approximately 3.4 percent and the housing price by 0.66 percent. The tax-and-grant package raises total births by approximately 4.9 percent and lowers the housing price by 18.80 percent. The central lesson is that housing policy affects family formation when it changes constrained households' access to family-sized space at the time they make fertility decisions.

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